

[Vitamin B12: Fact Sheet for Consumers \(original version\)](#)

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What is vitamin B12 and what does it do?

Vitamin B12 is a nutrient that helps keep your body's blood and nerve cells healthy and helps make DNA, the genetic material in all of your cells. Vitamin B12 also helps prevent megaloblastic anemia, a blood condition that makes people tired and weak.

How much vitamin B12 do I need?

The amount of vitamin B12 you need each day depends on your age. Average daily recommended amounts for different ages are listed below in micrograms (mcg).

Life Stage	Recommended Amount
Birth to 6 months	0.4 mcg
Infants 7–12 months	0.5 mcg
Children 1–3 years	0.9 mcg
Children 4–8 years	1.2 mcg
Children 9–13 years	1.8 mcg
Teens 14–18 years	2.4 mcg
Adults	2.4 mcg
Pregnant teens and women	2.6 mcg
Breastfeeding teens and women	2.8 mcg

What foods provide vitamin B12?

Vitamin B12 is found in a wide variety of animal foods. Plant foods have no vitamin B12 unless they are fortified. You can get recommended amounts of vitamin B12 by eating a variety of foods including the following:

- Fish, meat, poultry, eggs, milk, and other dairy products contain vitamin B12.

- Clams, oysters, and beef liver are some of the best sources of vitamin B12.
- Some breakfast cereals, nutritional yeasts, and other food products are fortified with vitamin B12.

To find out if a food has *added* vitamin B12, check the [Nutrition Facts label](#). Manufacturers are not required to list vitamin B12 on the label if a food naturally contains this vitamin.

What kinds of vitamin B12 dietary supplements are available?

Vitamin B12 is available in multivitamin/mineral supplements, in B-complex supplements, and in supplements containing only vitamin B12. It is usually in a form called cyanocobalamin. Other common forms are adenosylcobalamin, methylcobalamin, and hydroxycobalamin. Vitamin B12 is also available in a form that's dissolved under your tongue (called sublingual vitamin B12). Research has not shown that any form of supplemental vitamin B12 is better than the others.

The amount of vitamin B12 in supplements varies widely. Some provide doses of vitamin B12 that are much higher than recommended amounts, such as 500 mcg or 1,000 mcg, but your body absorbs only a small percentage of it. These doses are considered safe. Check the Supplement Facts label to see how much vitamin B12 a supplement contains.

A prescription form of vitamin B12 can be given as a shot. This is usually used to treat vitamin B12 deficiency. Vitamin B12 is also available by prescription as a nasal gel that's sprayed into the nose.

Am I getting enough vitamin B12?

Most people in the United States get enough vitamin B12 from the foods they eat. However, some people have trouble absorbing vitamin B12 from food. The body absorbs vitamin B12 from food in a two-step process. First, hydrochloric acid in the stomach separates vitamin B12 from the protein that it's attached to. Second, the freed vitamin B12 then combines with a protein made by the stomach, called intrinsic factor, and the body absorbs them together.

Vitamin B12 in dietary supplements isn't attached to protein and doesn't require the first step. However, B12 in supplements does need to combine with intrinsic factor to be absorbed.

People with pernicious anemia, an autoimmune disease, can't make intrinsic factor. As a result, they have trouble absorbing vitamin B12 from foods and dietary supplements.

Vitamin B12 deficiency affects between 3% and 43% of older adults. Your doctor can test your vitamin B12 level to see if you have a deficiency.

Certain groups of people may not get enough vitamin B12 or have trouble absorbing it:

- Many older adults don't have enough hydrochloric acid in their stomach to absorb the vitamin B12 that's naturally present in food. People over 50 should get most of their vitamin B12 from fortified foods or dietary supplements because, in most cases, their bodies can absorb vitamin B12 from these sources.
- People with an autoimmune disease called atrophic gastritis might not absorb enough vitamin B12 because they make too little hydrochloric acid and intrinsic factor in their stomach.
- People with pernicious anemia do not make the intrinsic factor needed to absorb vitamin B12. As a result, they have trouble absorbing vitamin B12 from foods and dietary supplements. Doctors usually treat pernicious anemia with vitamin B12 shots, although very high doses of vitamin B12 given by mouth might also be effective.
- People who have had some types of stomach or intestinal surgery (for example, to lose weight or to remove part or all of the stomach) might not make enough hydrochloric acid and intrinsic factor to absorb vitamin B12.
- People with disorders of the stomach and small intestine, such as celiac disease or Crohn's disease, might not absorb enough vitamin B12.

- People who eat little or no animal foods, such as vegetarians and vegans, might not get enough vitamin B12 from their diets. Only animal foods have vitamin B12 naturally. When pregnant women and women who breastfeed their babies are strict vegetarians or vegans, their babies might also not get enough vitamin B12.

What happens if I don't get enough vitamin B12?

Your body stores 1,000 to 2,000 times as much vitamin B12 as you'd typically eat in a day, so the symptoms of vitamin B12 deficiency can take several years to appear.

If you have a vitamin B12 deficiency, you may feel tired or weak. These are symptoms of megaloblastic anemia, which is a hallmark of vitamin B12 deficiency. You might also have pale skin, heart palpitations, loss of appetite, weight loss, and infertility. Your hands and feet might become numb or tingly, a sign of nerve problems. Other symptoms of vitamin B12 deficiency include problems with balance, depression, confusion, dementia, poor memory, and soreness of the mouth or tongue.

In infants, signs of a vitamin B12 deficiency include failure to thrive, delays in reaching the typical developmental milestones, and megaloblastic anemia.

Vitamin B12 deficiency can damage the nervous system even in people who don't have megaloblastic anemia, so it's important to treat a deficiency as soon as possible.

What are some effects of vitamin B12 on health?

Scientists are studying vitamin B12 to understand how it affects health. Here are several examples of what this research has shown.

Cancer

Some research shows that people with high levels of vitamin B12 have a higher risk of cancer. However, other research shows that the risk of cancer is higher in people with low levels of vitamin B12 or that vitamin B12 levels don't affect cancer risk. More evidence is needed to understand whether vitamin B12 levels affect cancer risk.

Heart disease and stroke

Vitamin B12 supplements (along with other B vitamins) reduce blood levels of homocysteine, a compound linked to an increased risk of having a heart attack or stroke. However, despite reducing homocysteine, research shows that these vitamins don't reduce the risk of developing cardiovascular disease or stroke.

Dementia and cognitive function

Most studies show that low blood levels of vitamin B12 don't affect the risk of cognitive decline in older people, regardless of whether they have dementia or Alzheimer's disease. More clinical trials are needed to better understand the effects of vitamin B12 supplementation on cognitive function in older adults.

Energy and endurance

Manufacturers often promote vitamin B12 supplements for energy, athletic performance, and endurance. However, vitamin B12 doesn't provide these benefits in people who get enough B12 from their diet.

Can vitamin B12 be harmful?

Vitamin B12 has not been shown to cause any harm, even at high doses.

Does vitamin B12 interact with medications or other dietary supplements?

Yes. Vitamin B12 supplements can interact or interfere with some medicines that you take. Here are several examples.

Gastric acid inhibitors

People take gastric acid inhibitors to treat certain digestion problems, such as gastroesophageal reflux disease and peptic ulcer disease. These drugs can interfere with vitamin B12 absorption from food by slowing the release of hydrochloric acid into the stomach, leading to vitamin B12 deficiency. Gastric acid inhibitors include omeprazole (Prilosec), lansoprazole (Prevacid), cimetidine (Tagamet), and ranitidine (Zantac).

Metformin

Metformin is used to treat prediabetes and diabetes. Metformin might reduce vitamin B12 absorption and lower blood levels of vitamin B12.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines. They can also explain whether the medicines you take might interfere with how your body absorbs or uses other nutrients.

Vitamin B12 and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about vitamin B12?

- For general information on vitamin B12
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Vitamin B12](#)
 - [Vitamin B12](#), MedlinePlus
- For more information on food sources of vitamin B12
 - USDA's [FoodData Central](#)
 - Nutrient Lists for vitamin B12 (listed by [food](#) or by [vitamin B12 content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

Alzheimer's disease

A brain disease in which thinking, memory, and reasoning ability is slowly destroyed. In advanced stages, an affected person becomes disoriented and confused, has mood and behavior changes, and has difficulty talking, walking, and swallowing. Alzheimer's disease is progressive, irreversible, and incurable.

anemia

A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

atrophic gastritis

A long-lasting (chronic) condition in which the lining of the stomach is inflamed. Gradually the lining wastes away, destroying the glands that make stomach acid.

autoimmune disease

A condition in which the body recognizes its own tissues as foreign and directs an immune response against them.

B vitamin

A nutrient that is important for cell function. The B vitamins are biotin, folate, niacin, pantothenic acid, riboflavin (vitamin B₂), thiamin, vitamin B₆, and vitamin B₁₂. The B vitamins make up the vitamin B complex.

body stores

The amount of a nutrient that stays in the body after eating and is available for future use. The size and location of this extra supply differs depending on the nutrient. For example, iron is stored in the liver.

cardiovascular disease

CVD. A general term referring to disorders of the heart and blood vessels. CVD includes coronary artery disease, heart failure, atherosclerosis, high blood pressure, peripheral artery disease, and stroke.

celiac disease

An autoimmune disorder in which eating gluten (a protein found in wheat, rye, barley, and possibly oats) causes the immune system to damage the small intestine, making it unable to absorb nutrients. It is a genetic disease that sometimes becomes active for the first time after surgery, pregnancy, childbirth, viral infection, or extreme stress. Also called sprue.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

clinical trial

A type of research study that uses volunteers to test the safety and efficacy (the ability to produce a beneficial effect) of new methods of screening (checking for disease when there are no symptoms), prevention, diagnosis, or treatment of a disease. Also called a clinical study.

cognitive function

Mental awareness and judgment.

compound

In pharmacy, a substance that contains more than one ingredient.

Crohn's disease

A long-lasting (chronic) disease that causes severe irritation in the gastrointestinal tract. It usually affects the lower small intestine (called the ileum) or the colon, but it can affect any part of the digestive tract from the mouth to the anus. It is painful, causing severe watery or bloody diarrhea, and may lead to life-threatening complications. Crohn's disease is a form of inflammatory bowel disease.

dairy food

Milk and products made with milk, such as buttermilk, yogurt, cheese, cottage cheese, and ice cream.

deficiency

An amount that is not enough; a shortage.

dementia

Damaged brain function (thinking, learning, making decisions, remembering) that worsens over time. It disrupts activities of daily living such as bathing, dressing, and walking.

deoxyribonucleic acid

DNA. The molecules inside cells that carry genetic information and pass it from one generation to the next.

depression

A serious medical condition that can interfere with an individual's ability to work, study, sleep, and eat. Symptoms include ongoing feelings of sadness and despair, loss of energy, and loss of interest or pleasure in activities that were once enjoyed. A person who has depression may experience changes in eating or sleeping habits, and have thoughts of death or suicide.

diabetes

A disease in which blood sugar (glucose) levels are high because the body is unable to use glucose properly. Diabetes occurs when the body does not make enough insulin, which helps the cells use glucose, or when the body no longer responds to insulin.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

digestion

The process the body uses to break down food into simple substances for energy, growth, and cell repair.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

evidence

Information used to support the use of a particular screening procedure, treatment, or preventive measure. In medicine, evidence needed to determine effectiveness is provided by laboratory research, clinical trials, and other studies.

failure to thrive

A condition in which infants and children are dramatically smaller or shorter than other children of the same age and gender, and physical, mental, and social skills are significantly delayed. Causes include medical disorders, environmental factors, malnutrition, and neglect.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gastric

Having to do with the stomach.

gastroesophageal reflux disease

GERD. A condition in which stomach acid leaks back into the esophagus because the muscle between the stomach and the esophagus does not close properly. It causes frequent heartburn and can lead to more serious health problems such as ulcers, swallowing difficulties, and cancer.

gene

The functional and physical unit of heredity passed from parent to offspring. Genes are pieces of DNA, and most genes contain the information for making a specific protein.

heart attack

The blockage of an artery supplying blood and oxygen to the heart, resulting in the damage or death of a section of heart muscle.

heart palpitation

Forceful and irregular beating of the heart.

homocysteine

An amino acid (a building block of protein). At high blood levels, it may increase the risk of developing coronary heart disease, heart attack, stroke, and Alzheimer's disease. Elevated homocysteine may also increase the risk of developing osteoporosis and bone fractures.

hydrochloric acid

An acid made in the stomach. It works with enzymes (substances that speed up chemical reactions in the body) to break down proteins during digestion.

infant

A child younger than 12 months old.

infertility

The inability to produce children.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

intestine

The section of the digestive tract below the stomach, including the small and large intestines, rectum, and anus.

intrinsic factor

A protein made by the stomach that is needed to absorb vitamin B₁₂ in the large intestine.

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

megaloblastic anemia

A disorder in which red blood cells are larger than normal, immature, and few in number, which reduces the amount of oxygen that can be carried by the blood to the body's tissues. It is caused by a deficiency in folate or vitamin B₁₂.

microgram

μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nasal

Having to do with the nose.

nerve

A bundle of microscopic fibers that carries messages back and forth from the brain to other parts of the body.

nervous system

The brain and spinal cord, including the network of nerves that carry messages back and forth between the brain and all parts of the body. The nervous system controls what the body does.

neuron

A nerve cell. Neurons send chemical and electrical messages throughout the nervous system that direct the body to function, move, think, and have emotions.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

nutritional yeast

A food product or food additive made from yeast (a fungus). The yeast is pasteurized (heated) to prevent it from growing in a person's digestive tract. Nutritional yeast is used as a source of protein and B vitamins. Some (but not all) brands of nutritional yeast contain vitamin B₁₂. Latin name:

Saccharomyces cerevisiae.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by

evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

peptic ulcer disease

A sore or hole in the lining of the stomach or the first part of the small intestine, causing burning pain in the gut. Most ulcers are caused by an infection with a type of bacteria called *Helicobacter pylori* (*H. pylori*); other causes include long-term use of nonsteroidal anti-inflammatory drugs (such as aspirin and ibuprofen), alcohol, and tobacco.

pernicious anemia

An autoimmune disease that prevents the body from making intrinsic factor (a protein made by the stomach and needed to absorb vitamin B12 in the intestine). If left untreated, pernicious anemia causes vitamin B12 deficiency which leads to megaloblastic anemia (a disorder in which red blood cells are larger than normal, immature, and few in number, which reduces the amount of oxygen that can be carried by the blood to the body's tissues).

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

sign

An indication of disease that can be seen and/or measured. Examples include high fever, high blood pressure, infection, and coughing up blood.

small intestine

The part of the digestive tract that is located between the stomach and the large intestine.

stroke

A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time.

<http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm>

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vegan

A person who eats only plant-based foods. Vegans do not eat meat, poultry, fish, eggs, milk, dairy products, or honey, and do not use leather, silk, or wool, or soaps and cosmetics that are made from animal products.

vegetarianism

The practice of avoiding all or most animal products for environmental, philosophical, and health reasons. Vegetarians (people who practice vegetarianism) eat a diet based on foods that come from plants and may include some dairy products and eggs. See: vegetarian diet.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin B₁₂

A group of chemical compounds that contain cobalt and are needed for certain chemical reactions in the body. Vitamin B₁₂ is involved in maintaining healthy nerve cells and red blood cells. It is needed to make DNA (the genetic material in all cells), and is required for the metabolism (chemical changes that take place in the tissues to produce energy and the basic materials needed by the body) of carbohydrate, fat, and protein. Also called cobalamin. For more information see the NIH Office of Dietary Supplements Vitamin B₁₂ fact sheet.

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